

CSE 2106: Microprocessor and Assembly Lab



Department of Computer Science and Engineering, University of Dhaka

Lab 4

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Outline

- 1 [Your Task](#)
- 2 [Submission Guideline](#)

3

4

5

6

7

8

9

10

Your Task

- Write an ARM Assembly language program to perform division on 8 bit, 16 bit and 32 bit numbers.
- Write an ARM Assembly language program to convert a BCD number in memory to its equivalent HEX number
- Write an ARM Assembly language program to convert a decimal number to its equivalent binary number.
- Write an ARM Assembly language program to calculate **the checksum** of a block of N bytes stored in memory.
[The checksum is computed by **adding all bytes together (including carry)** and taking the **1's complement** of the final sum.]

Your Task

- Write an ARM Assembly Language Program that converts a given total number of seconds into **hours**, **minutes**, and **remaining seconds**. You will map register as follows: R0 = Total Seconds, R1 = Hours, R2 = Minutes, R3 = Seconds
- Write an ARM Assembly language program to reverse the bits of a 16-bit number. You will map register as follows: R0 = Input Number, R1 = Reversed number.

Submission Guideline

- ① Your Assembly code with proper comments. (*.s file)
- ② A document (*.tex file) that contains:
- ③ Detail explanation of the code
 - Screenshot that shows the state of the system after the code has been loaded.
 - Screenshot that shows the situation after the code has been executed.
 - Submit as a .zip file. Example: your classroll lab#.zip (12 lab2.zip)